

Conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

21 September 2026

Summary

- Goal: generate the post-spin-up (T, S) state of the Veros ACC channel directly from the EKE-closure coefficients (c_k, c_ϵ) , skipping spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble; runs end to end on Jean Zay in under one A100 hour.
- Result: with per-level normalisation the model’s hold-out error is $\approx$ the mean-state baseline (0.15 K); it beats the nearest training run only when the hold-out gap is large, never beats interpolation in $\log c_k$; a Wasserstein metric on the stratification profiles gives the same ranking and shows that training and held-out conditions score alike.
- Physical miss: the model generates temperature inversions at the rate of the weakly eddying runs in every regime (Table 1).

1 Data

- 100 Veros ACC runs, 10×10 log-spaced grid: $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$; 100 years, 30-day output, interior grid $30 \times 42 \times 15$ (x, y, z).
- Kept: T and S , last 20 years (241 snapshots/run). $S = 35$ psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; bottom level still drifts by 0.45 K over the window.
- One effective parameter: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean T ; the low- c_k/c_ϵ part of the grid saturates.

2 Method

- Fields $\rightarrow$ 30 channels $(T_{1..15}, S_{1..15})$, normalised per level with training statistics, land $\rightarrow 0$, zero-padded $42 \times 30 \rightarrow 48 \times 32$:

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

99.4 % of the training values lie in the sampler’s clip range $[-1, 1]$; the exceptions are the southern wall between 650 and 1400 m and the two bottom levels.

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition c , the standardised pair $(\log c_k, \log c_\epsilon)$, enters through an MLP added to the timestep embedding.
- DDPM, ϵ -prediction, 1000 steps, squared-cosine schedule, clipping, EMA:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c) \right\|^2. \quad (2)$$

20 000 steps, batch 32, AdamW 10^{-4} with cosine decay; 22 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; land and padding re-zeroed after every step; 32 samples per condition.

3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs; interpolation in $\log c_k$ between the nearest training rows below (b) and above (a) in the same c_ϵ column, $\hat{x} = (1 - w) \bar{x}_b + w \bar{x}_a$, $w = \frac{\log c_k - \log c_k^b}{\log c_k^a - \log c_k^b}$ (one-sided when a side is missing).
- Three hold-out splits, each with its own training set and statistics: scattered (10 interior points), band of three rows ($c_k = 0.126, 0.2, 0.3175$; 30 runs; training rows a factor 6.3 apart), top row ($c_k = 0.8$; 10 runs; extrapolation).
- Ensemble of n samples: $\text{RMSE}_n^2 \approx \text{bias}^2 + \text{pattern}^2 + \text{spread}^2 / n$; going from 8 to 32 samples changed the RMSE by 0.004 K only ($0.154 \rightarrow 0.150$), as predicted.
- Distributional metric. Each state is reduced to its profile of horizontal means over water cells, $\bar{T}_z$. Per level, the Wasserstein-1 distance between the 32 generated values and all 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with Δz_z the level thicknesses from the level midpoints ($H = 2080$ m). All 241 snapshots are used, not a subsample of 32: the quantile form accepts unequal sizes and a subsample would only add noise. A point prediction x (the baselines) gives $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$. Floor: 32 random true snapshots against all 241, mean over 100 draws (what a perfect generator with 32 samples would score); drift scale: first half of the window against the second. Unlike the RMSE of the ensemble mean, this rewards a correct spread and does not reward collapse to the mean; it ignores horizontal structure, which the RMSE keeps.

	scattered 10 runs	band of three rows 30 runs	top row 10 runs
diffusion, ensemble mean of 32: RMSE (K)	0.150	0.129	0.193
diffusion, single sample: RMSE (K)	0.186	0.175	0.232
interpolation in $\log c_k$: RMSE (K)	–	0.083	0.170
nearest training run: RMSE (K)	0.037	0.116	0.170
neighbour average: RMSE (K)	0.020	0.116	0.170
mean training state: RMSE (K)	0.155	0.197	0.470
diffusion: domain-mean bias (K)	−0.024	−0.023	−0.083
ensemble spread / true spread (K)	0.110 / 0.031	0.113 / 0.031	0.114 / 0.032
T inversions (interfaces), generated / truth	9.6 % / 7.7 %	11.0 % / 1.3 %	12.0 % / 0.1 %
max salinity error (psu)	0.011	0.012	0.020
W_1 profile, diffusion (K)	0.125	0.109	0.161
W_1 profile, interpolation in $\log c_k$ (K)	–	0.062	0.142
W_1 profile, nearest training run (K)	0.044	0.095	0.142
W_1 profile, floor (32 of 241 true) / drift (K)	0.007 / 0.066	0.007 / 0.067	0.008 / 0.073

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells only, 32 samples per run.

4 Findings

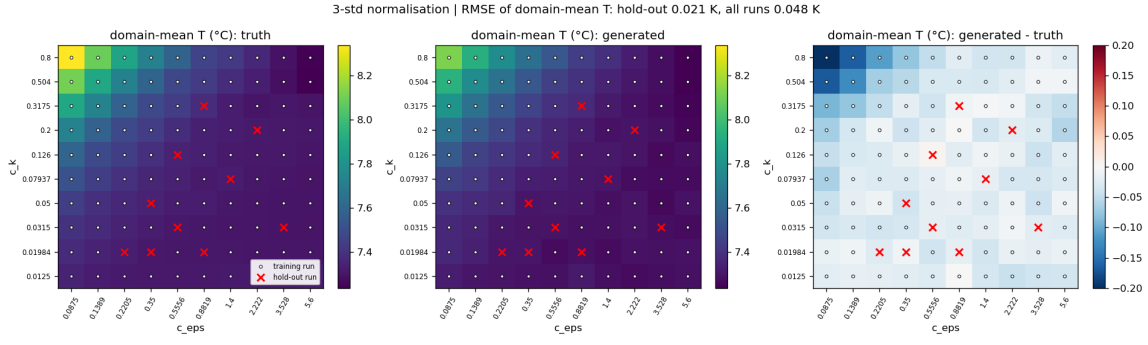
- **Conditioning is weak under per-level scaling.** Signal-to-range ratio $\rho_z = \sigma_z^{\text{runs}} / (3\sigma_z)$ is 0.7 % at the surface, 9 % at 650 m, 25 % at the bottom; the ϵ -loss weights all cells equally, so the shared structure dominates. Scattered split: RMSE = mean-state baseline (Figure 1).
- **Model error is split-independent (0.13–0.19 K); baselines degrade with the gap (0.02–0.17 K).** The model beats the nearest run in the two upper band rows and over the trending part of the top row (c_ϵ 0.22–0.88), never beats $\log c_k$ interpolation (Figures 3, 5).

- **Depth structure.** Best method at 650–800 m in the band split (0.08 K vs 0.11 K); worse than baselines in the top 250 m (scale $3\sigma_z \approx 12$ K amplifies sampler noise) and at the two bottom levels (skewed, drifting, clipped training distribution).
- **Extrapolation is bounded by clipping.** Under the top split’s training statistics, 17 % of the true bottom values of the held-out row exceed $[-1, 1]$; generated corner 0.2 K too cold, bottom RMSE 0.41 vs 0.22 K for the row below.
- **Inversions.** Generated 10–12 % of interfaces with $\partial_z T < 0$ in every regime; truth: 7.7 % (scattered) $\rightarrow$ 1.3 % (band) $\rightarrow$ 0.1 % (top). The regime dependence is not learned.
- **W_1 on the mean profiles agrees with the RMSE ranking and adds two facts** (Figures 2–6). The model sits at 0.11–0.16 K against a floor of 0.007 K and a drift scale of 0.07 K; the interpolation baseline is at 0.04–0.14 K. (i) Training grid points score the same as held-out ones (0.123 vs 0.125 K, scattered split): the model does not fit the training conditions either, the signature of the mean-state collapse. (ii) Between 650 and 1200 m the samples beat every baseline under W_1 in the band and top splits, because a point prediction is charged for the spread it cannot represent; the model’s total is set by the top 250 m and the bottom two levels, as for the RMSE.
- **Plumbing.** S within 0.02 psu of 35 with no constraint; land exact; ensemble spread $3.5\times$ the true within-window spread (Figures 7–9).

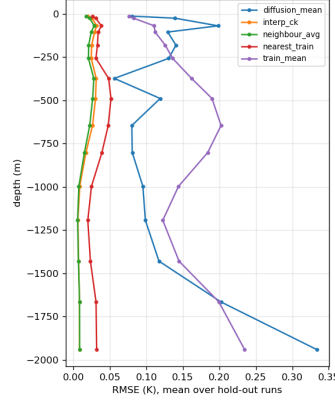
5 Next steps

- Stratification constraint at sampling time (DINO-Fusion’s isotonic projection, on T since S is constant): targets the inversion miss directly.
- Stronger conditioning: classifier-free guidance; year as a third condition (drift of 0.45 K at the bottom inside the window).
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread–skill).
- Larger clip range or scale factor where extrapolation matters.
- Then: same pipeline on a global Veros configuration (active S , seasonal cycle, multi-century spin-up).

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`; commits `f912e66` (scattered split) and `5991188` (band, top). Extraction 2.5 min (8 cores), training 22 min (one A100), generation + evaluation 3.5 min.

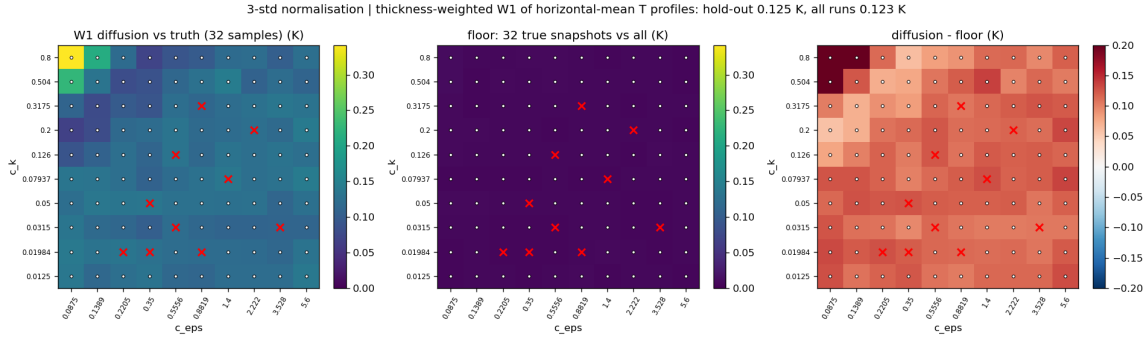


(a) Domain-mean T over the grid: truth, generated, difference. White dots: training runs; red crosses: held-out runs.

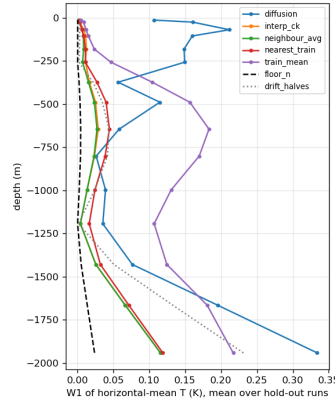


(b) RMSE by depth, mean over held-out runs.

Figure 1: Scattered split (10 interior held-out runs), 32 samples per run.

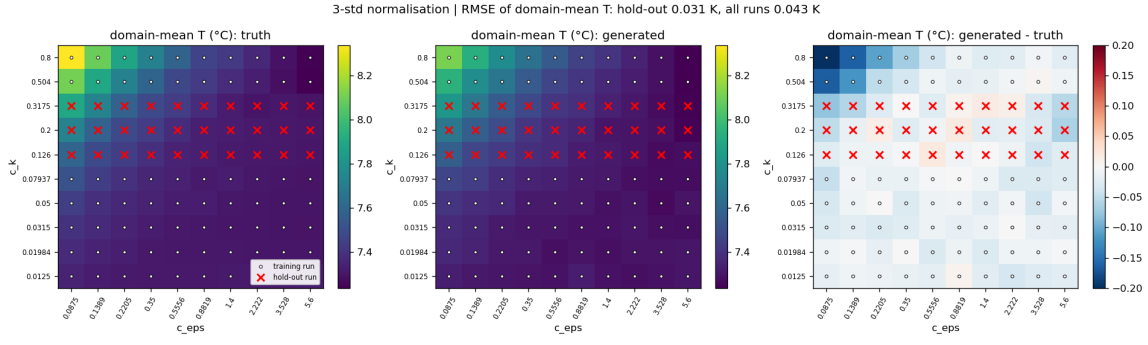


(a) Thickness-weighted W_1 of the horizontal-mean T profile over the grid: diffusion vs truth, floor, difference.

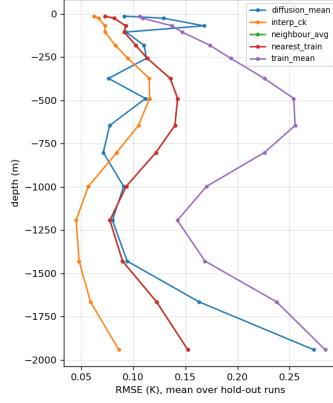


(b) W_1 by level; dashed: floor; dotted: drift.

Figure 2: W_1 metric, Scattered split (10 interior held-out runs), 32 samples per run.

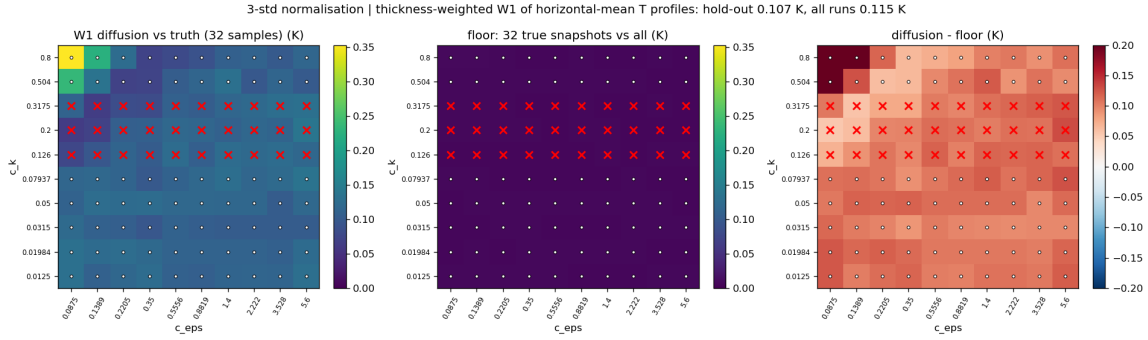


(a) Domain-mean T over the grid.

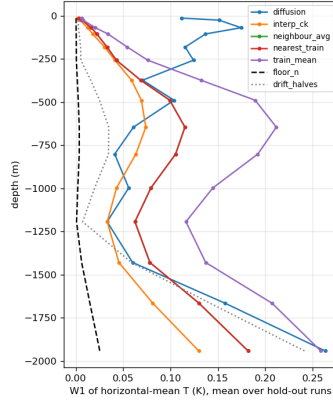


(b) RMSE by depth; orange: $\log c_k$ interpolation.

Figure 3: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.

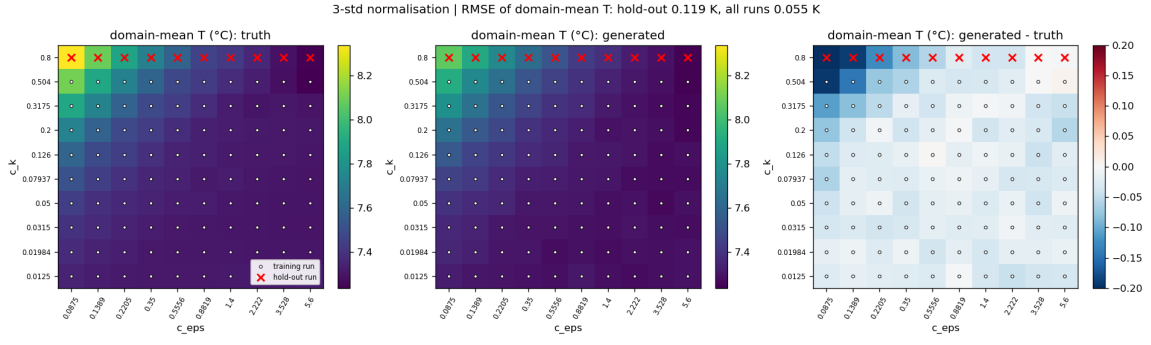


(a) Thickness-weighted W_1 of the horizontal-mean T profile over the grid: diffusion vs truth, floor, difference.

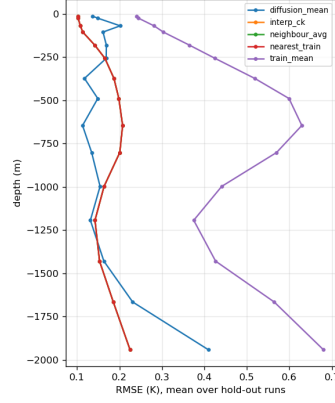


(b) W_1 by level; dashed: floor; dotted: drift.

Figure 4: W_1 metric, Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.

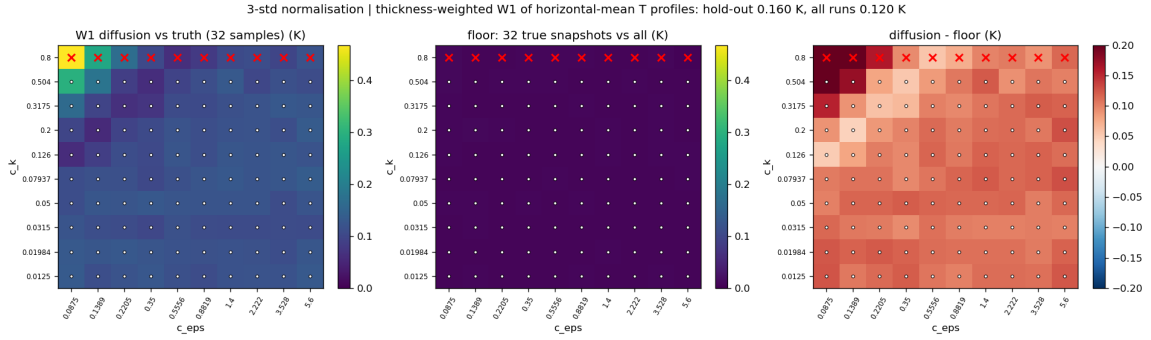


(a) Domain-mean T over the grid.

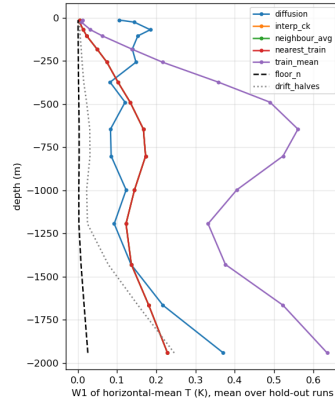


(b) RMSE by depth; the three baselines coincide (row below).

Figure 5: Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.



(a) Thickness-weighted W_1 of the horizontal-mean T profile over the grid: diffusion vs truth, floor, difference.



(b) W_1 by level; dashed: floor; dotted: drift.

Figure 6: W_1 metric, Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.

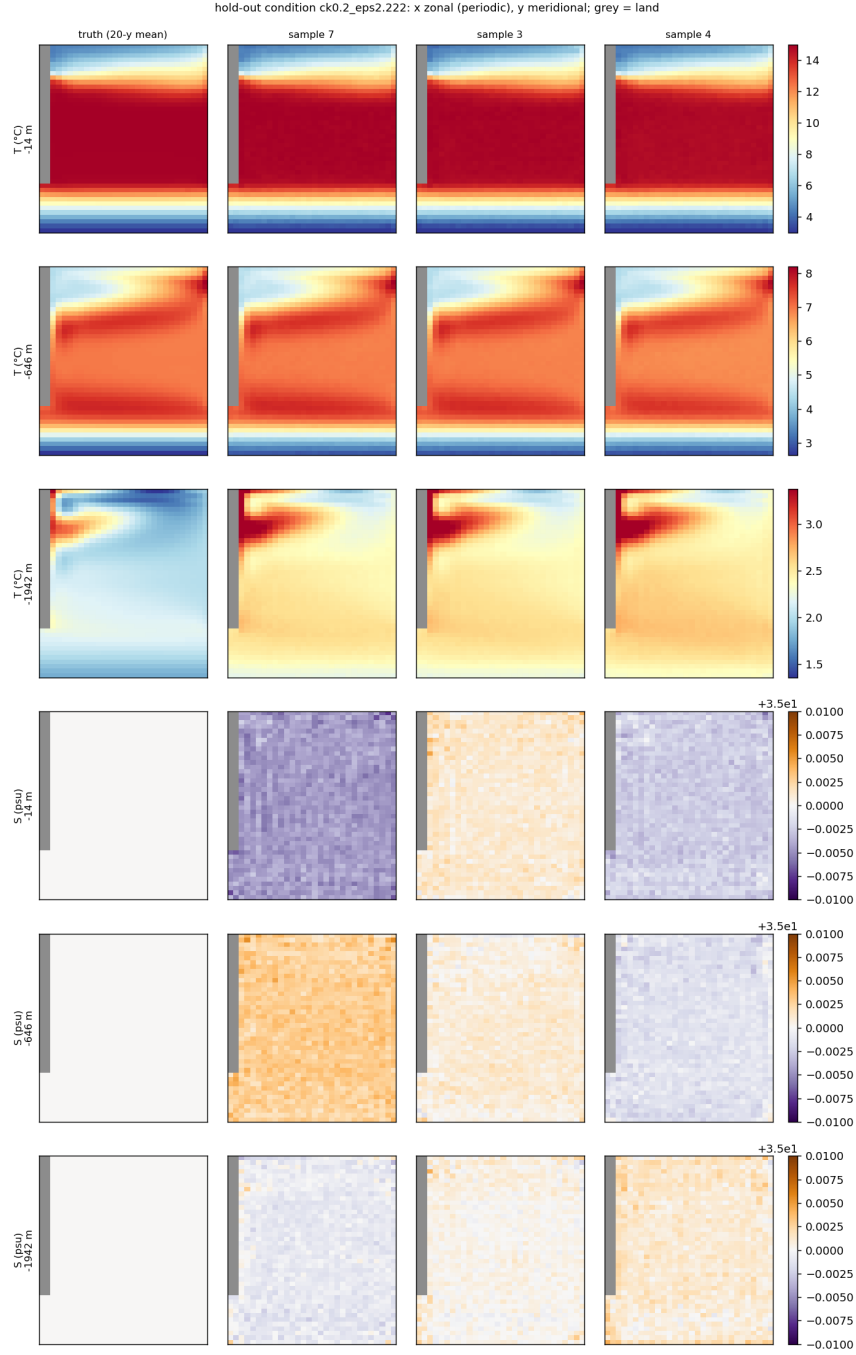


Figure 7: Scattered split, $c_k = 0.2$, $c_\epsilon = 2.222$: true 20-year mean (left) and three random samples at the surface, 650 m and the bottom; T (top three rows), S (bottom three rows). Grey: land.

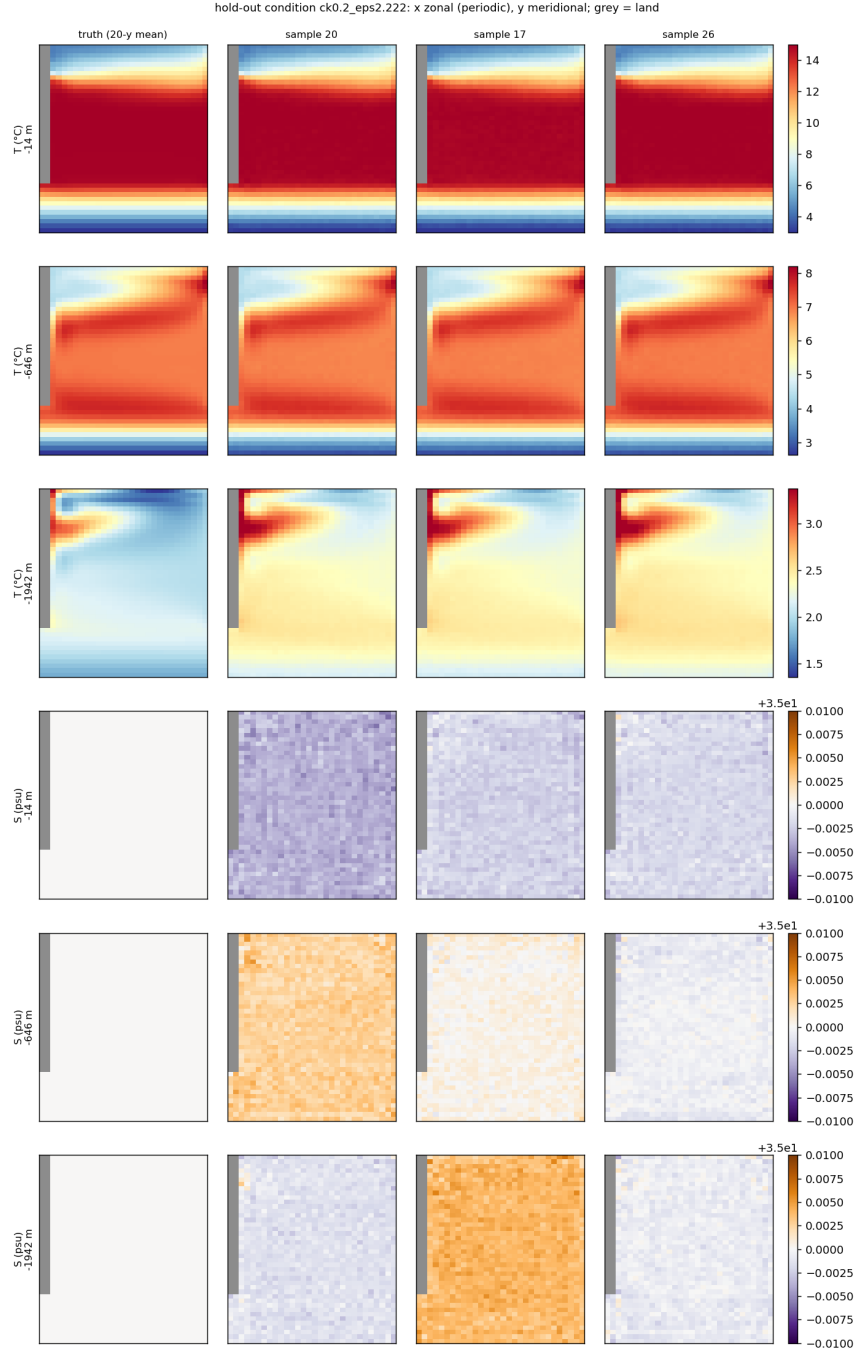


Figure 8: Band-of-three split, same condition ($c_k = 0.2$, $c_{\epsilon} = 2.222$, now in the middle held-out row).

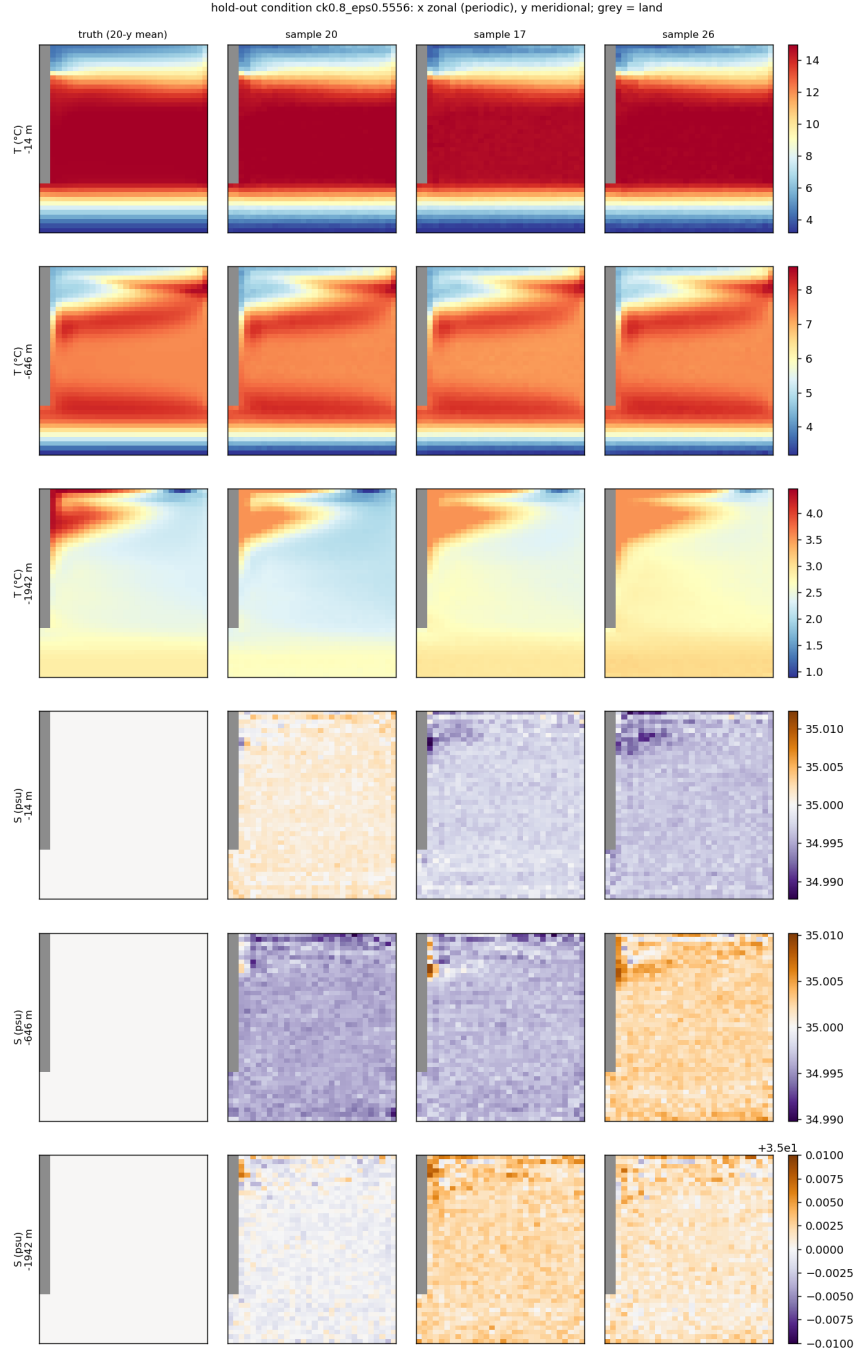


Figure 9: Top-row split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (extrapolation one row above the training grid).